



2361 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Cycle: 1, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	Henan MIRI	MIRI Medium Resolution Spectroscopy	(1) 2085HENAN
	3	Klumpkea MIRI	MIRI Medium Resolution Spectroscopy	(5) 1040KLUMPKEA
	4	Anastasia MIRI	MIRI Medium Resolution Spectroscopy	(2) 824ANASTASIA
	5	Hercynia MIRI	MIRI Medium Resolution Spectroscopy	(6) 458HERCYNIA
	6	Haremar MIRI	MIRI Medium Resolution Spectroscopy	(7) 1372HAREMARI
	2	Henan NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(1) 2085HENAN
	7	Klumpkea NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(5) 1040KLUMPKEA
	8	Anastasia NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(2) 824ANASTASIA
	9	Hercynia NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(6) 458HERCYNIA
	109	Hercynia NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(6) 458HERCYNIA
	209	Hercynia NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(6) 458HERCYNIA
	10	Haremar NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(7) 1372HAREMARI

ABSTRACT

Spectroscopic and polarimetric studies of a subset of the L-type asteroids has indicated that these objects may contain ~30% calcium-aluminum inclusions (CAIs), the first solids to condense out of the disk during the Solar System's formation. This is a surprising result because even the most

CAI-abundant meteorites contain only 3-5% of these inclusions. High CAI content is considered to be indicator of ancient origin, and the confirmation of the CAI abundance in these asteroids would make them plausibly the most ancient objects in the Solar System. In addition, their existence has important implications for the timing of events in the Solar System's early history and the origins of the Solar System's raw materials. The spinel that is a major constituent of CAIs has prominent spectral signatures in the mid-IR, and mid-IR spectroscopy would readily detect this material and robustly differentiate high-CAI bodies from other candidate asteroid compositions. We propose to observe a set of L-type asteroids with high apparent CAI content in order to confirm or refute the high-CAI hypothesis, and to constrain the disk formation region and thermal/aqueous alteration histories of these unusual objects.

OBSERVING DESCRIPTION

We will use MIRI/MRS and NIRSPEC to obtain 2.5-28 micron spectra of five L-Type asteroids that have high purported CAI fractions.

Proposal 2361 - Targets - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Solar System Targets	#	Name	Level 1	Level 2	Level 3
	(1)	2085HENAN	TYPE=ASTEROID,A=2.69948759798329,E=0.08463 589801153751,I=3.835582070121222,O=118.1696001 232119,W=294.9281900159813,M=39.038884466452 71,EQUINOX=J2000,EPOCH=04-OCT- 2014:00:00:00,EpochTimeScale=TDB		
	Comments: Extended=NO				
	(2)	824ANASTASIA	TYPE=ASTEROID,A=2.794227296484664,E=0.1358 856549573977,I=8.122637904548894,O=141.4074111 017484,W=141.9617267970302,M=201.74284494159 91,EQUINOX=J2000,EPOCH=04-JUN- 2013:00:00:00,EpochTimeScale=TDB		
	Comments: Extended=NO				
	(5)	1040KLUMPKEA	TYPE=ASTEROID,A=3.110218907007033,E=0.1934 723385533078,I=16.69395729756977,O=280.1855936 059181,W=158.016260062192,M=78.1335989309848 9,EQUINOX=J2000,EPOCH=11-NOV- 2013:00:00:00,EpochTimeScale=TDB		
	Comments: Extended=NO				
	(6)	458HERCYNIA	TYPE=ASTEROID,A=2.998675826348398,E=0.2373 876324674879,I=12.66789786328738,O=134.5707567 719593,W=275.8897180930277,M=139.44043488609 62,EQUINOX=J2000,EPOCH=28-NOV- 2011:00:00:00,EpochTimeScale=TDB		
	Comments: Extended=NO				
	(7)	1372HAREMARI	TYPE=ASTEROID,A=2.766783482735815,E=0.1460 825503216309,I=16.45644482158624,O=327.4848470 307606,W=88.57402005460411,M=352.69102607487 67,EQUINOX=J2000,EPOCH=19-JUN- 2014:00:00:00,EpochTimeScale=TDB		
Comments: Extended=NO					

Proposal 2361 - Observation 1 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Observation	Proposal 2361, Observation 1: Henan MIRI													Wed Aug 09 17:02:15 GMT 2023
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
	(Henan MIRI (Obs 1)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.													
Solar System Targets	#	Name	Level 1					Level 2				Level 3		
	(1)	2085HENAN	TYPE=ASTEROID,A=2.69948759798329,E=0.08463 589801153751,I=3.835582070121222,O=118.1696001 232119,W=294.9281900159813,M=39.038884466452 71,EQUINOX=J2000,EPOCH=04-OCT- 2014:00:00:00,EpochTimeScale=TDB											
	Comments: Extended=NO													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction			
	F560W	ALL				YES			FULL		NEUTRAL			
Dithers	#	Dither Type					Optimized For			Direction				
	1	4-Point					POINT SOURCE			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wbk.Calc ID	
	1		IMAGER	F1000W	FASTR1	8	12	1	Dither 1	4	48	1187.717		
	1	LONG(C)	MRSLONG		FASTR1	8	12	1	Dither 1	4	48	1187.717		
	1	LONG(C)	MRSSHORT		FASTR1	8	12	1	Dither 1	4	48	1187.717		
	2		IMAGER	F1000W	FASTR1	8	12	1	Dither 1	4	48	1187.717		
	2	MEDIUM(B)	MRSLONG		FASTR1	8	12	1	Dither 1	4	48	1187.717		
	2	MEDIUM(B)	MRSSHORT		FASTR1	8	12	1	Dither 1	4	48	1187.717		
	3		IMAGER	F1000W	FASTR1	8	12	1	Dither 1	4	48	1187.717		
	3	SHORT(A)	MRSLONG		FASTR1	8	12	1	Dither 1	4	48	1187.717	63543.8	
	3	SHORT(A)	MRSSHORT		FASTR1	8	12	1	Dither 1	4	48	1187.717		

Proposal 2361 - Observation 1 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Special Requirements	DEFAULT WINDOW: ANGULAR RATE 2085HENAN FROM JWST LESS THAN 0.03
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Proposal 2361 - Observation 3 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Observation	Proposal 2361, Observation 3: Klumpkea MIRI												Wed Aug 09 17:02:15 GMT 2023
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(Klumpkea MIRI (Obs 3)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Solar System Targets	#	Name	Level 1					Level 2				Level 3	
	(5)	1040KLUMPKEA	TYPE=ASTEROID,A=3.110218907007033,E=0.1934 723385533078,I=16.69395729756977,O=280.1855936 059181,W=158.016260062192,M=78.1335989309848 9,EQUINOX=J2000,EPOCH=11-NOV- 2013:00:00:00,EpochTimeScale=TDB										
	Comments: Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F560W	ALL				NO			FULL		NEUTRAL		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					POINT SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	SHORT(A)	MRSLONG		FASTR1	8	30	1	Dither 1	4	120	2985.943	63543.9
	1	SHORT(A)	MRSSHORT		FASTR1	8	30	1	Dither 1	4	120	2985.943	
	2	MEDIUM(B)	MRSLONG		FASTR1	8	30	1	Dither 1	4	120	2985.943	
	2	MEDIUM(B)	MRSSHORT		FASTR1	8	30	1	Dither 1	4	120	2985.943	
	3	LONG(C)	MRSLONG		FASTR1	8	30	1	Dither 1	4	120	2985.943	
	3	LONG(C)	MRSSHORT		FASTR1	8	30	1	Dither 1	4	120	2985.943	

Proposal 2361 - Observation 3 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Special Requirements	DEFAULT WINDOW: ANGULAR RATE 1040KLUMPKEA FROM JWST LESS THAN 0.03
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Proposal 2361 - Observation 4 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Observation	Proposal 2361, Observation 4: Anastasia MIRI												Wed Aug 09 17:02:15 GMT 2023		
	Diagnostic Status: Warning														
	Observing Template: MIRI Medium Resolution Spectroscopy														
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.														
	(Anastasia MIRI (Obs 4)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.														
Solar System Targets	#	Name	Level 1				Level 2				Level 3				
	(2)	824ANASTASIA	TYPE=ASTEROID,A=2.794227296484664,E=0.1358 856549573977,I=8.122637904548894,O=141.4074111 017484,W=141.9617267970302,M=201.74284494159 91,EQUINOX=J2000,EPOCH=04-JUN- 2013:00:00:00,EpochTimeScale=TDB												
	Comments: Extended=NO														
Acquisition	#	Target													
	1	NONE													
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray				Grating Wheel Direction	
	F560W	ALL				YES				FULL				NEUTRAL	
Dithers	#	Dither Type				Optimized For				Direction					
	1	4-Point				POINT SOURCE				NEGATIVE					
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID		
	1		IMAGER	F1000W	FASTR1	5	8	1	Dither 1	4	32	521.708	63543.12		
	1	LONG(C)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708			
	1	LONG(C)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708			
	2		IMAGER	F1000W	FASTR1	5	8	1	Dither 1	4	32	521.708	63543.12		
	2	MEDIUM(B)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708			
	2	MEDIUM(B)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708			
	3		IMAGER	F1000W	FASTR1	5	8	1	Dither 1	4	32	521.708	63543.12		
	3	SHORT(A)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708	63543.7		
	3	SHORT(A)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708			

Proposal 2361 - Observation 4 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Special Requirements	DEFAULT WINDOW: ANGULAR RATE 824ANASTASIA FROM JWST LESS THAN 0.03
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Proposal 2361 - Observation 5 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Observation	Proposal 2361, Observation 5: Hercynia MIRI												Wed Aug 09 17:02:15 GMT 2023	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
	(Hercynia MIRI (Obs 5)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.													
Solar System Targets	#	Name	Level 1				Level 2				Level 3			
	(6)	458HERCYNIA	TYPE=ASTEROID,A=2.998675826348398,E=0.2373 876324674879,I=12.66789786328738,O=134.5707567 719593,W=275.8897180930277,M=139.44043488609 62,EQUINOX=J2000,EPOCH=28-NOV- 2011:00:00:00,EpochTimeScale=TDB											
	Comments: Extended=NO													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
	F560W	ALL				YES				SUB64		NEUTRAL		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wbk.Calc ID	
	1		IMAGER	F1000W	FASTR1	5	40	1	Dither 1	4	160	81.375	63543.11	
	1	LONG(C)	MRSLONG		FASTR1	5	40	1	Dither 1	4	160	2652.938		
	1	LONG(C)	MRSSSHORT		FASTR1	5	40	1	Dither 1	4	160	2652.938		
	2		IMAGER	F1000W	FASTR1	5	40	1	Dither 1	4	160	81.375	63543.11	
	2	MEDIUM(B)	MRSLONG		FASTR1	5	40	1	Dither 1	4	160	2652.938		
	2	MEDIUM(B)	MRSSSHORT		FASTR1	5	40	1	Dither 1	4	160	2652.938		
	3		IMAGER	F1000W	FASTR1	5	40	1	Dither 1	4	160	81.375	63543.11	
	3	SHORT(A)	MRSLONG		FASTR1	5	40	1	Dither 1	4	160	2652.938		
	3	SHORT(A)	MRSSSHORT		FASTR1	5	40	1	Dither 1	4	160	2652.938	63543.6	

Proposal 2361 - Observation 5 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Special Requirements	DEFAULT WINDOW: ANGULAR RATE 458HERCYNIA FROM JWST LESS THAN 0.03
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Proposal 2361 - Observation 6 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Observation	Proposal 2361, Observation 6: Hare mari MIRI												Wed Aug 09 17:02:15 GMT 2023
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(Hare mari MIRI (Obs 6)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Solar System Targets	#	Name	Level 1					Level 2			Level 3		
	(7)	1372HAREMARI	TYPE=ASTEROID,A=2.766783482735815,E=0.1460 825503216309,I=16.45644482158624,O=327.4848470 307606,W=88.57402005460411,M=352.69102607487 67,EQUINOX=J2000,EPOCH=19-JUN-2014:00:00:00,EpochTimeScale=TDB										
	Comments: Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F560W	ALL				YES			FULL		NEUTRAL		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					POINT SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1		IMAGER	F1000W	FASTR1	5	16	1	Dither 1	4	64	1054.515	63543.10
	1	LONG(C)	MRSLONG		FASTR1	5	16	1	Dither 1	4	64	1054.515	
	1	LONG(C)	MRSSSHORT		FASTR1	5	16	1	Dither 1	4	64	1054.515	
	2		IMAGER	F1000W	FASTR1	5	16	1	Dither 1	4	64	1054.515	63543.10
	2	MEDIUM(B)	MRSLONG		FASTR1	5	16	1	Dither 1	4	64	1054.515	
	2	MEDIUM(B)	MRSSSHORT		FASTR1	5	16	1	Dither 1	4	64	1054.515	
	3		IMAGER	F1000W	FASTR1	5	16	1	Dither 1	4	64	1054.515	63543.10
	3	SHORT(A)	MRSLONG		FASTR1	5	16	1	Dither 1	4	64	1054.515	63532.5
	3	SHORT(A)	MRSSSHORT		FASTR1	5	16	1	Dither 1	4	64	1054.515	

Proposal 2361 - Observation 6 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Special Requirements	DEFAULT WINDOW: ANGULAR RATE 1372HAREMARI FROM JWST LESS THAN 0.03
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Proposal 2361 - Observation 2 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Observation	Proposal 2361, Observation 2: Henan NIRSPEC										Wed Aug 09 17:02:15 GMT 2023	
	Diagnostic Status: Warning											
Observing Template: NIRSpec Fixed Slit Spectroscopy												
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Henan NIRSPEC (Obs 2)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1			Level 2			Level 3			
	(1)	2085HENAN	TYPE=ASTEROID,A=2.69948759798329,E=0.08463 589801153751,I=3.835582070121222,O=118.1696001 232119,W=294.9281900159813,M=39.038884466452 71,EQUINOX=J2000,EPOCH=04-OCT- 2014:00:00:00,EpochTimeScale=TDB									
Comments: Extended=NO												
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	63522.1	
Template	Slit					Subarray						
	S200A1					SUBS200A1						
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	5						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235M/F170LP	S200A1	NRS	10	4	1	NONE	5	20	1277.97	
	2	G395M/F290LP	S200A1	NRS	10	4	2	NONE	5	20	1277.97	63522.9

Proposal 2361 - Observation 2 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Special Requirements	DEFAULT WINDOW: ANGULAR RATE 2085HENAN FROM JWST LESS THAN 0.03
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Proposal 2361 - Observation 7 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Observation	Proposal 2361, Observation 7: Klumpkea NIRSPEC										Wed Aug 09 17:02:15 GMT 2023	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Klumpkea NIRSPEC (Obs 7)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2				Level 3	
	(5)	1040KLUMPKEA	TYPE=ASTEROID,A=3.110218907007033,E=0.1934 723385533078,I=16.69395729756977,O=280.1855936 059181,W=158.016260062192,M=78.1335989309848 9,EQUINOX=J2000,EPOCH=11-NOV- 2013:00:00:00,EpochTimeScale=TDB									
	Comments: Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	63522.5	
Template	Slit					Subarray						
	S200A1					SUBS200A1						
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	5						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235M/F170LP	S200A1	NRS	10	4	1	NONE	5	20	1277.97	
	2	G395M/F290LP	S200A1	NRS	10	4	2	NONE	5	20	1277.97	63522.10

Proposal 2361 - Observation 7 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Special Requirements	DEFAULT WINDOW: ANGULAR RATE 1040KLUMPKEA FROM JWST LESS THAN 0.03
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Proposal 2361 - Observation 8 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Observation	Proposal 2361, Observation 8: Anastasia NIRSPEC										Wed Aug 09 17:02:15 GMT 2023	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Anastasia NIRSPEC (Obs 8)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2				Level 3	
	(2)	824ANASTASIA	TYPE=ASTEROID,A=2.794227296484664,E=0.1358 856549573977,I=8.122637904548894,O=141.4074111 017484,W=141.9617267970302,M=201.74284494159 91,EQUINOX=J2000,EPOCH=04-JUN- 2013:00:00:00,EpochTimeScale=TDB									
	Comments: Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	63522.2	
Template	Slit					Subarray						
	S200A1					SUBS200A1						
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235M/F170LP	S200A1	NRS	10	3	1	NONE	3	9	575.086	
	2	G395M/F290LP	S200A1	NRS	10	3	2	NONE	3	9	575.086	63522.7

Proposal 2361 - Observation 8 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Special Requirements	DEFAULT WINDOW: ANGULAR RATE 824ANASTASIA FROM JWST LESS THAN 0.03
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Proposal 2361 - Observation 9 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Observation	Proposal 2361, Observation 9: Hercynia NIRSPEC										Wed Aug 09 17:02:15 GMT 2023	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Hercynia NIRSPEC (Obs 9)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1			Level 2			Level 3			
	(6)	458HERCYNIA	TYPE=ASTEROID,A=2.998675826348398,E=0.2373 876324674879,I=12.66789786328738,O=134.5707567 719593,W=275.8897180930277,M=139.44043488609 62,EQUINOX=J2000,EPOCH=28-NOV- 2011:00:00:00,EpochTimeScale=TDB									
	Comments: Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	63522.3	
Template	Slit					Subarray						
	S200A1					SUBS200A1						
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	5						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235M/F170LP	S200A1	NRS	10	2	1	NONE	5	10	638.985	
	2	G395M/F290LP	S200A1	NRS	10	2	2	NONE	5	10	638.985	63522.6

Proposal 2361 - Observation 9 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Special Requirements	DEFAULT WINDOW: ANGULAR RATE 458HERCYNIA FROM JWST LESS THAN 0.03
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Proposal 2361 - Observation 109 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Observation	Proposal 2361, Observation 109: Hercynia NIRSPEC										Wed Aug 09 17:02:15 GMT 2023	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 109:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Hercynia NIRSPEC (Obs 109)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2				Level 3	
	(6)	458HERCYNIA	TYPE=ASTEROID,A=2.998675826348398,E=0.2373 876324674879,I=12.66789786328738,O=134.5707567 719593,W=275.8897180930277,M=139.44043488609 62,EQUINOX=J2000,EPOCH=28-NOV- 2011:00:00:00,EpochTimeScale=TDB									
	Comments: Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	63522.3	
Template	Slit					Subarray						
	S200A1					SUBS200A1						
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	5						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235M/F170LP	S200A1	NRS	10	2	1	NONE	5	10	638.985	
	2	G395M/F290LP	S200A1	NRS	10	2	2	NONE	5	10	638.985	63522.6

Proposal 2361 - Observation 109 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Special Requirements	DEFAULT WINDOW: ANGULAR RATE 458HERCYNIA FROM JWST LESS THAN 0.03
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Proposal 2361 - Observation 209 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Observation	Proposal 2361, Observation 209: Hercynia NIRSPEC										Wed Aug 09 17:02:15 GMT 2023	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 209:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Hercynia NIRSPEC (Obs 209)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1			Level 2			Level 3			
	(6)	458HERCYNIA	TYPE=ASTEROID,A=2.998675826348398,E=0.2373 876324674879,I=12.66789786328738,O=134.5707567 719593,W=275.8897180930277,M=139.44043488609 62,EQUINOX=J2000,EPOCH=28-NOV- 2011:00:00:00,EpochTimeScale=TDB									
	Comments: Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	63522.3	
Template	Slit					Subarray						
	S200A1					SUBS200A1						
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	5						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235M/F170LP	S200A1	NRS	10	2	1	NONE	5	10	638.985	
	2	G395M/F290LP	S200A1	NRS	10	2	2	NONE	5	10	638.985	63522.6

Proposal 2361 - Observation 209 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Special Requirements	DEFAULT WINDOW: ANGULAR RATE 458HERCYNIA FROM JWST LESS THAN 0.03
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Proposal 2361 - Observation 10 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Observation	Proposal 2361, Observation 10: Haremari NIRSPEC										Wed Aug 09 17:02:15 GMT 2023	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Haremari NIRSPEC (Obs 10)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2				Level 3	
	(7)	1372HAREMARI	TYPE=ASTEROID,A=2.766783482735815,E=0.1460 825503216309,I=16.45644482158624,O=327.4848470 307606,W=88.57402005460411,M=352.69102607487 67,EQUINOX=J2000,EPOCH=19-JUN- 2014:00:00:00,EpochTimeScale=TDB									
	Comments: Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	63522.4	
Template	Slit					Subarray						
	S200A1					SUBS200A1						
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	5						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235M/F170LP	S200A1	NRS	10	4	1	NONE	5	20	1277.97	
	2	G395M/F290LP	S200A1	NRS	10	4	2	NONE	5	20	1277.97	63522.8

Proposal 2361 - Observation 10 - Are L-type asteroids ancient reservoirs for the earliest Solar System solids?

Special Requirements	DEFAULT WINDOW: ANGULAR RATE 1372HAREMARI FROM JWST LESS THAN 0.03
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